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COMPUTATIONAL MODEL

Effects of Optimization Technique on Simulated Muscle Activations and Forces

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Two optimization techniques, static optimization (SO) and computed muscle control (CMC), are often used in OpenSim to estimate the muscle activations and forces responsible for movement. Although differences between SO and CMC muscle function have been reported, the accuracy of each technique and the combined effect of optimization and model choice on simulated muscle function is unclear. The purpose of this study was to quantitatively compare the SO and CMC estimates of muscle activations and forces during gait with the experimental data in the Gait2392 and Full Body Running models. In OpenSim (version 3.1), muscle function during gait was estimated using SO and CMC in 6 subjects in each model and validated against experimental muscle activations and joint torques. Experimental and simulated activation agreement was sensitive to optimization technique for the soleus and tibialis anterior. Knee extension torque error was greater with CMC than SO. Muscle forces, activations, and co-contraction indices tended to be higher with CMC and more sensitive to model choice. CMC's inclusion of passive muscle forces, muscle activation-contraction dynamics, and a proportional-derivative controller to track kinematics contributes to these differences. Model and optimization technique choices should be validated using experimental activations collected simultaneously with the data used to generate the simulation.

Keywords: musculoskeletal models, dynamic simulations, OpenSim, static optimization, computed muscle control

Musculoskeletal modeling and simulation techniques enable estimates of variables that influence movement, including muscle activations and forces. Many of these variables are not commonly determined in human experiments because of the significant pain and discomfort to the subject with approaches such as fine-wire electromyography (EMG) and tendon buckle transducers. OpenSim[1] is an open-source musculoskeletal modeling and simulation software employed by over 50,000 unique users[2] to investigate dynamic movements such as walking,[3–5] running,[6,7] rising from a chair,[8] and climbing stairs.[9]

OpenSim allows users several choices for developing simulations of movement. Several musculoskeletal models are compatible with OpenSim to study similar movements; therefore, users must choose a model with an understanding of the implications of the differences between models.[10] Several studies have investigated the relative impact of the differences in the model parameters on the simulation results.[10–14] Recent work determined that differences in joint and segment coordinate system definitions and differences in muscle parameters between 3-dimensional musculoskeletal models affected simulated joint mechanics and muscle function.[10–12]

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In addition to model selection, users must choose an optimization technique to estimate muscle activations and forces. OpenSim includes 2 optimization techniques, static optimization (SO)[15] and computed muscle control (CMC),[16,17] which apply different methods to estimate the muscle activations and forces that would reproduce the experimental motion. Given the known kinematic state of the model at each time point, SO resolves the net joint moments into individual muscle forces subject to a criterion that minimizes the sum of the squared muscle activations. The SO algorithm computes the active force along a muscle's tendon, assuming a rigid tendon and neglecting the contributions of passive muscle forces. The CMC algorithm also solves a static optimization to determine the muscle excitations that will achieve the desired accelerations to track the experimental motion. In SO, the known experimental accelerations are the desired accelerations. However, CMC also includes a proportional derivative control law to account for error between the current model state and the experimental kinematics. In addition, CMC accounts for temporal delays between muscle excitation and force development by using a forward integration of the muscle activation-contraction dynamics to determine a feasible range of muscle forces that could be produced at the subsequent time point. The CMC algorithm also includes the passive muscle force contribution to the steady-state muscle force.[17] The muscle excitations determined from the CMC's static optimization are input into a forward simulation that determines the current kinematic state and desired accelerations for the next time step.

Several studies have compared the simulated muscle function between SO and CMC,[18–23] and some have suggested that SO is the superior optimization technique for estimating muscle function in human locomotion due to its robustness and computational efficiency.[22,23] However, joint torques determined from CMC activations using the forward integration of muscle contraction dynamics

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more accurately reproduce inverse dynamics (ID) joint torques compared with those from SO, which has been attributed to the inclusion of muscle activation-contraction dynamics in the CMC algorithm.[18] Yet, other studies investigating the effect of optimization techniques found that, compared with SO, CMC estimated a larger sum of muscle forces for gait and the sit-to-stand transfer in humans[20] and larger muscle activations and forces during walking and running in an ostrich model.[21] The authors of these 2 studies suggest that CMC estimates larger activations and forces because it accounts for passive muscle dynamics and activation-contraction dynamics.[20,21] This hypothesis has been supported by a recent study that demonstrated that large muscle coactivations estimated by CMC can be attributed to excessive passive muscle forces that cause compensatory forces from antagonist muscles that are not reflected in the experimental activation patterns.[24] In addition, one study cited the forward integration step of CMC as the cause of larger variations in muscle forces estimated by CMC than in SO when the body segment parameters were adjusted to ±40% of their nominal value in increments of 10%, suggesting differences in musculoskeletal models may have a greater effect on muscle force and activation estimated by CMC than by SO.[19]

Although the aforementioned studies have examined differences between the SO and CMC estimates of muscle function, these studies focused on muscle function at a single joint,[20,23] of a single muscle,[19] or a single model.[18,21,22] Given the important role of optimization techniques in uncovering the underlying muscle function responsible for movement and the known differences between musculoskeletal models in muscle function estimated by the same optimization technique,[10] there is a need to quantify the accuracy of the SO and CMC estimates of muscle forces and activations and to evaluate the combined effect of different musculoskeletal models and optimization techniques on simulated muscle function during gait. Therefore, the purpose of this study was to quantitatively compare the SO and CMC estimates of muscle activations and forces during gait to the experimental data using OpenSim and 2 associated musculoskeletal models: Gait2392[1] and the Full Body Running (Hamner) model.[6] These 2 models have the same muscles, muscle parameters, and pelvis neutral position, but Hamner includes representations of the arms, resulting in 2 additional segments and 6 additional degrees of freedom compared with Gait2392,[6] which could lead to significant differences in the simulated muscle forces and activations using the same optimization algorithm, as demonstrated by Roelker et al.[10]

Methods

The kinematic, kinetic, and EMG data of 6 healthy young adult subjects (4 female and 2 male; age = 21 [2.3] y; weight = 69.1 [8.3] kg; height = 1.70 [0.05] m) walking at their self-selected speed (1.30 [0.14] $m \cdot s^{-1}$) were collected in a previously described study.[5] After providing written informed consent in accordance with the institutional review board of The Ohio State University, the subjects completed 5 overground walking trials at a self-selected speed, while the position of the reflective markers placed on the body according to the Full Body Point-Cluster Technique[25] were collected at 150 Hz using an 8-camera Vicon MX-F40 system (Vicon Motion Systems, Oxford, UK). Ground reaction forces were simultaneously collected from 6 force plates (Bertec Corp, Columbus, OH) at 1500 Hz. Bilateral surface EMG (Noraxon, Scottsdale, AZ) was collected at 1500 Hz on the long head of the biceps femoris, gluteus maximus, gluteus medius, medial gastrocnemius, rectus femoris, soleus, tibialis anterior, and vastus lateralis. The EMG was high-pass filtered at 10 Hz, rectified, and root mean square (RMS) smoothed, with a 20 millisecond window. The EMG was normalized to the peak EMG activation in the gait cycle (GC) such that the normalized EMG ranged from 0 to 1.

Musculoskeletal Models and Simulations

As previously described,[10] OpenSim[1] (version 3.1; OpenSim, Stanford University, Stanford, CA) was used to simulate one GC for each subject in Gait2392 and Hamner. For each subject, the segment dimensions of Gait2392 and Hamner were scaled using the relative distance between pairs of experimental markers to adjust the location of the corresponding virtual markers in the model.[1] The scaling procedure results in the same whole body mass for a given subject's Gait2392 and Hamner model. Segment masses and inertial properties are also identical between models for a subject, with the exception of the torso mass, because Gait2392's torso mass is equal to the sum of the masses of Hamner's torso and left and right humerus, ulna, radius, and hand. Then, inverse kinematics was used to estimate the joint angles that best reproduced the experimental marker data, and a residual reduction algorithm reduced dynamic inconsistencies.[1] The residual reduction algorithm solutions for joint kinematics and kinetics did not significantly differ between Gait2392 and Hamner, with the exception of an average 3° greater ankle range of motion during stance with Gait2392 than with Hamner.[10] The difference in kinematics occurred due to the addition of arm segments in Hamner and the greater number of markers necessary to track the arm segments, which alters the weighted least squares problem solved in the inverse kinematics step.[10] The set of muscle activations and forces that would reproduce each subject's gait were determined using SO and CMC, with muscle activations bounded between 0 (not activated) and 1 (fully activated) in SO and between 0.02 and 1 in CMC. For each subject, the same residual reduction algorithm kinematics and external loads were used as inputs into both optimization techniques for a model. The default parameters (eg, maximum number of integrator steps, maximum integration step size) provided by OpenSim were used in SO and CMC. The parameter values are identical across both optimization techniques, with the exception of the maximum number of optimization iterations (100 for SO and 1000 for CMC). In addition, the same lower-extremity actuator parameters were used across all simulations. The Thelen 2003 Muscle Model[26] was used in both musculoskeletal models. The maximum residual forces, moments, and reserves were required to be less than 25 N, 75 N·m, and 50 N·m, respectively.[27]

Analysis

The simulated muscle activations and forces were compared between the 4 simulation conditions (2 models × 2 optimization techniques): Gait2392-SO, Gait2392-CMC, Hamner-SO, and Hamner-CMC. For each subject in each condition, muscle activations and forces were analyzed for 10 muscles/muscle groups: biceps femoris long head, gastrocnemius (medial and lateral head), gluteus maximus, gluteus medius, iliacus, medial hamstrings (semimembranosus and semitendinosus), rectus femoris, soleus, tibialis anterior, and vasti (vastus intermedius, vastus medialis, and vastus lateralis). For each muscle in a muscle group (eg, gastrocnemius), the muscle's simulated activation profile was weighted by the ratio of the muscle's peak isometric force to the sum of the muscle group's peak isometric forces:

$$w_i = \frac{F_i^{\max}}{\sum_{i=1}^{m} F_i^{\max}}, \quad (1)$$

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where the weight (w) of muscle i is calculated as muscle i's peak isometric force ($F_i^{\max}$) divided by the sum of the peak isometric forces of the m muscles in the group. The weights from Equation 1 were used to calculate the activation of the muscle group (a_{Group}; Equation 2) as the weighted sum of activations of the m muscles in the group:

$$a_{\text{Group}} = \sum_{i=1}^{m} w_i \times a_i \tag{2}$$

The simulated force profile of a muscle group was calculated as the sum of the force profiles of the muscles in the group. All further analyses of muscle group forces and activations were performed using the muscle groups' aggregate profiles.

The effect of the optimization technique and model choice on the validity of the simulated muscle activations and forces estimated by each condition was evaluated in accordance with suggested best practices.[28] To assess the agreement between experimental and simulated activation patterns, RMS errors and the cosine of similarity (COS) were calculated between each condition's simulated muscle activation patterns and the normalized EMG. The RMS error was chosen because it is a common measure of the difference between the estimated and observed values and accounts for differences in magnitudes between 2 curves. The COS was also chosen as a metric of agreement between simulated and experimental activation patterns because it compares the orientation of 2 vectors independent of their magnitude, thereby evaluating the agreement in timing between the activation patterns without the confounding effect of differences in magnitude. The COS values can range from −1 to 1, with 1 indicating that the 2 vectors have the same orientation, 0 indicating a 90° offset in orientation, and −1 indicating a 180° offset in orientation. Thus, a COS closer to 1 indicates better agreement in activation timing between simulated and experimental patterns. In addition, joint torques produced by CMC and SO forces were calculated by multiplying the simulated forces by the muscles' moment arms over the GC and compared with joint torques derived by ID. The agreement between ID- and simulation-derived joint torques was quantified by RMS error. For both the RMS and COS analyses, the SO and CMC activations were normalized to the maximum SO and CMC activation in the simulation, respectively, such that the simulated activations had a maximum value of 1 for each muscle.

To compare the simulation results between conditions with respect to both the magnitude and timing of the estimated muscle activations and forces simultaneously, peak activations and forces were determined for 3 phases of the GC: early stance (0%–30% GC), late stance (30%–65%), and swing (65%–100%). For each condition, the average peak activation and force of each muscle were calculated in each phase across all subjects.

Finally, to analyze the simulated muscle activations in a clinically relevant manner, muscle cocontraction indices (CCIs) were calculated[29,30] for 5 muscle pairs for each model using each optimization technique. The CCIs of the lateral vasti and hamstrings (VLLH), medial vasti and hamstrings (VMMH), lateral vasti and gastrocnemius (VLLG), and medial vasti and gastrocnemius (VMMG) were calculated during weight acceptance (0%–15% GC) because increased coactivation of these muscles is a compensation strategy for quadriceps weakness exhibited by older adults and knee osteoarthritis patients.[31] The calculation of tibialis anterior and soleus CCIs was performed during early midstance (15%–30% GC) because tibialis anterior and soleus coactivation is a balance control strategy observed in older adults[32] and diabetic neuropathy patients.[33] Each CCI is calculated as the integrated sum of the cocontraction (Equation 3)[29] of the muscle, with the lower activation (lower ACT) and the muscle in the higher activation (higher ACT) at each point in the gait phase:

$$\text{CCI} = \int_{i=1}^{i=n} \frac{\text{lower ACT}_i}{\text{higher ACT}_i} \times (\text{lower ACT}_i + \text{higher ACT}_i), \tag{3}$$

where i is the time point in the phase.

Statistics

Separate 2-way repeated-measures analyses of variance assessed the main fixed effects of the model and optimization technique and the interaction effect of the model-x-optimization technique on RMS errors, COS, peak muscle activations and forces for each GC phase, and CCIs. Tukey post hoc tests were used to assess for pairwise differences between conditions. Statistical analyses were performed in Minitab Statistical Software (version 16.2.4; Minitab, Inc, State College, PA), with a significance level of $\alpha = .05$ set a priori.

Results

The agreement between experimental and simulated activation pattern magnitude, quantified by RMS error (Table 1, Figure 1), was similar between conditions for all muscles ($P \geq .056$), with the exception of tibialis anterior, for which the RMS errors were greater with CMC than with SO ($P = .033$). The average RMS errors ranged from 0.465 (largest error; gluteus medius with Hamner-SO) to 0.263 (smallest error; tibialis anterior with Hamner-SO). There were no significant differences in the average RMS errors across all muscles between conditions (see "Condition average" in Table 1; $P \geq .499$).

The agreement between experimental and simulated activation pattern timing, quantified by COS (Table 2, Figure 1), was not significantly different between conditions for all muscles ($P \geq .057$), with the exception of soleus, for which COS was greater with Hamner-CMC than all other conditions ($P \leq .016$), greater with CMC than SO ($P = .027$), and greater with Hamner than Gait2392 ($P = .032$). The average COS ranged from 0.428 (largest error; soleus with Gait2392-SO) to 0.731 (smallest error; tibialis anterior with Gait2392-CMC). There were no significant differences in average COS across all muscles between conditions (see "Condition average" in Table 2; $P \geq .201$).

All conditions generated joint torques that closely matched the ID-derived joint torques, with average RMS errors of <7 N·m across all joint torques and conditions (Figure 2). However, there were statistically significant differences in errors between conditions. The RMS error between ID- and simulation-derived knee extension torque was significantly greater with CMC than SO ($P = .022$). Hip adduction RMS errors were greater with Hamner than Gait2392 ($P = .049$), while ankle dorsiflexion RMS errors were greater with Gait2392 than Hamner ($P = .002$). There were no significant differences in RMS errors in hip flexion or internal rotation torques between conditions ($P \geq .08$).

Peak SO muscle activations and forces were similar between models; however, peak CMC muscle activations and forces differed between models and differed from peak SO activations and forces within a model (Figures 3–5). The primary differences

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Table 1 Average (SD) RMS Error by Muscle for Each Model-Optimization Technique Condition

	Gait2392		Hamner		
Muscle	**SO**	**CMC**	**SO**	**CMC**	**Muscle average**
Gluteus maximus	0.343 (0.071)	0.321 (0.065)	0.328 (0.073)	0.340 (0.068)	0.333 (0.065)
Gluteus medius	0.430 (0.059)	0.440 (0.053)	0.465 (0.051)	0.433 (0.061)	0.442 (0.054)
Rectus femoris	0.367 (0.055)	0.332 (0.030)	0.361 (0.068)	0.335 (0.034)	0.349 (0.049)
Vastus lateralis	0.293 (0.048)	0.316 (0.041)	0.298 (0.045)	0.287 (0.040)	0.298 (0.042)
Biceps femoris	0.264 (0.053)	0.286 (0.069)	0.268 (0.057)	0.344 (0.056)	0.291 (0.064)
Medial gastrocnemius	0.348 (0.012)	0.363 (0.021)	0.346 (0.020)	0.338 (0.108)	0.349 (0.053)
Soleus	0.330 (0.043)	0.339 (0.036)	0.326 (0.037)	0.295 (0.028)	0.323 (0.038)
Tibialis anterior*	0.289 (0.075)	0.355 (0.058)	0.263 (0.068)	0.306 (0.019)	0.303 (0.065)
Condition average	0.333 (0.071)	0.344 (0.063)	0.332 (0.079)	0.335 (0.069)	

Abbreviations: CMC, computed muscle control; RMS, root mean square; SO, static optimization. Notes: Values closer to 0 indicate better agreement with experimental data. *Indicates statistically significant difference ($P < .05$) between optimization techniques.

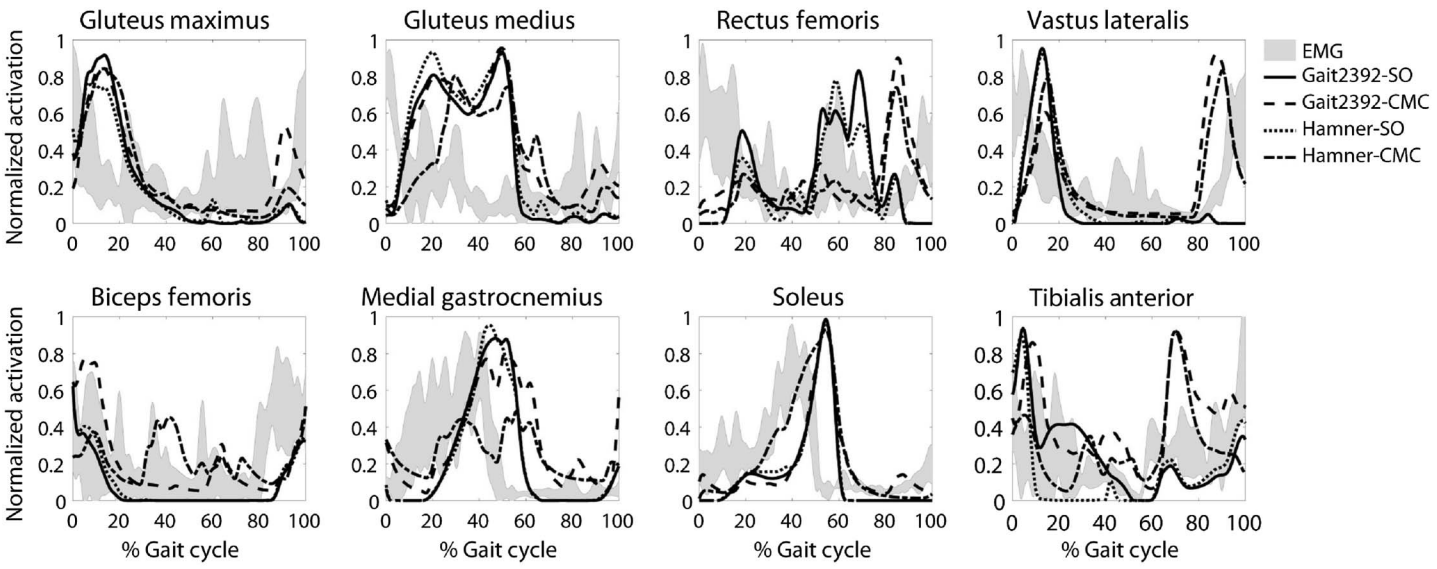

Figure 1 — Experimental EMG (shaded area; average ± 1 SD) and average simulated activation patterns for Gait2392-SO (solid line), Gait2392-CMC (dashed line), Hamner-SO (dotted line), and Hamner-CMC (dotted-dashed line). Experimental and simulated activations were normalized to the peak activation in the respective trial. SDs for individual condition activation patterns can be found in the Appendix Figures A1–A4. CMC indicates computed muscle control; EMG, electromyography; SO, static optimization.

between conditions are described here, and a thorough discussion of the individual differences in peak muscle activations and forces between conditions is reported in the Appendix. Of the 30 peak activation comparisons (10 muscles × 3 phases) between conditions, 18 were significantly different between models, optimization techniques, and/or conditions ($P \leq .021$). Peak activations estimated by CMC were significantly greater than those of SO for 16 comparisons, while SO activations were greater than those of CMC for only one comparison (early stance biceps femoris long head activation). Among all 30 comparisons, the average difference between the conditions with the maximum and minimum mean peak activation was 0.228 (0.179) (91.1% [56.0%] difference). Of the 30 peak force comparisons between conditions, 20 were significantly different between models, optimization techniques, and/or conditions ($P \leq .044$). The peak forces estimated by CMC were significantly greater than those of SO for 13 comparisons, while the SO forces were greater than those of CMC for only one comparison (early stance biceps femoris long head force). Among all 30 comparisons, the average difference between the conditions with the maximum and minimum mean peak force was 292.8 (183.3) N (87.0% [51.5%] difference).

Differences in the muscle activations between conditions resulted in differences in the CCIs computed from simulated muscle activations (Figure 6). Greater CCIs were calculated from CMC than from SO for VMMH ($P = .025$), VLLG ($P = .004$), and VMMG ($P = .003$). Specifically, Gait2392-CMC produced the largest VLLG ($P \leq .001$) and VMMG ($P \leq .003$) CCIs compared with all other conditions. Hamner-SO's tibialis anterior and soleus CCI was less than that of all other conditions ($P \leq .043$). There were no significant differences between conditions for lateral vasti and hamstrings ($P \geq .053$).

Discussion

This study aimed to determine how different musculoskeletal model and optimization technique combinations influenced

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Table 2 Average (SD) Cosine of Similarity by Muscle for Each Model-Optimization Technique Condition

	Gait2392		Hamner		
Muscle	**SO**	**CMC**	**SO**	**CMC**	**Muscle average**
Gluteus maximus	0.517 (0.142)	0.608 (0.124)	0.523 (0.113)	0.552 (0.130)	0.550 (0.125)
Gluteus medius	0.512 (0.076)	0.579 (0.081)	0.539 (0.085)	0.462 (0.105)	0.523 (0.092)
Rectus femoris	0.477 (0.147)	0.504 (0.128)	0.451 (0.184)	0.454 (0.110)	0.471 (0.137)
Vastus lateralis	0.492 (0.163)	0.564 (0.103)	0.516 (0.144)	0.623 (0.125)	0.549 (0.137)
Biceps femoris	0.560 (0.195)	0.561 (0.222)	0.553 (0.212)	0.430 (0.167)	0.526 (0.195)
Medial gastrocnemius	0.548 (0.093)	0.551 (0.089)	0.564 (0.057)	0.449 (0.388)	0.528 (0.198)
Soleus*,**,***	0.428 (0.131)	0.432 (0.101)	0.446 (0.122)	0.662 (0.091)	0.492 (0.145)
Tibialis anterior	0.625 (0.135)	0.731 (0.108)	0.602 (0.118)	0.647 (0.060)	0.651 (0.113)
Condition average	0.520 (0.141)	0.566 (0.142)	0.524 (0.137)	0.535 (0.187)	

Abbreviations: CMC, computed muscle control; SO, static optimization. Notes: Values closer to 1 indicate better agreement with experimental data. Symbols indicate statistically significant differences ($P < .05$) between models (*), optimization technique (**), and pairwise differences between conditions (***). See text for detailed description of pairwise differences.

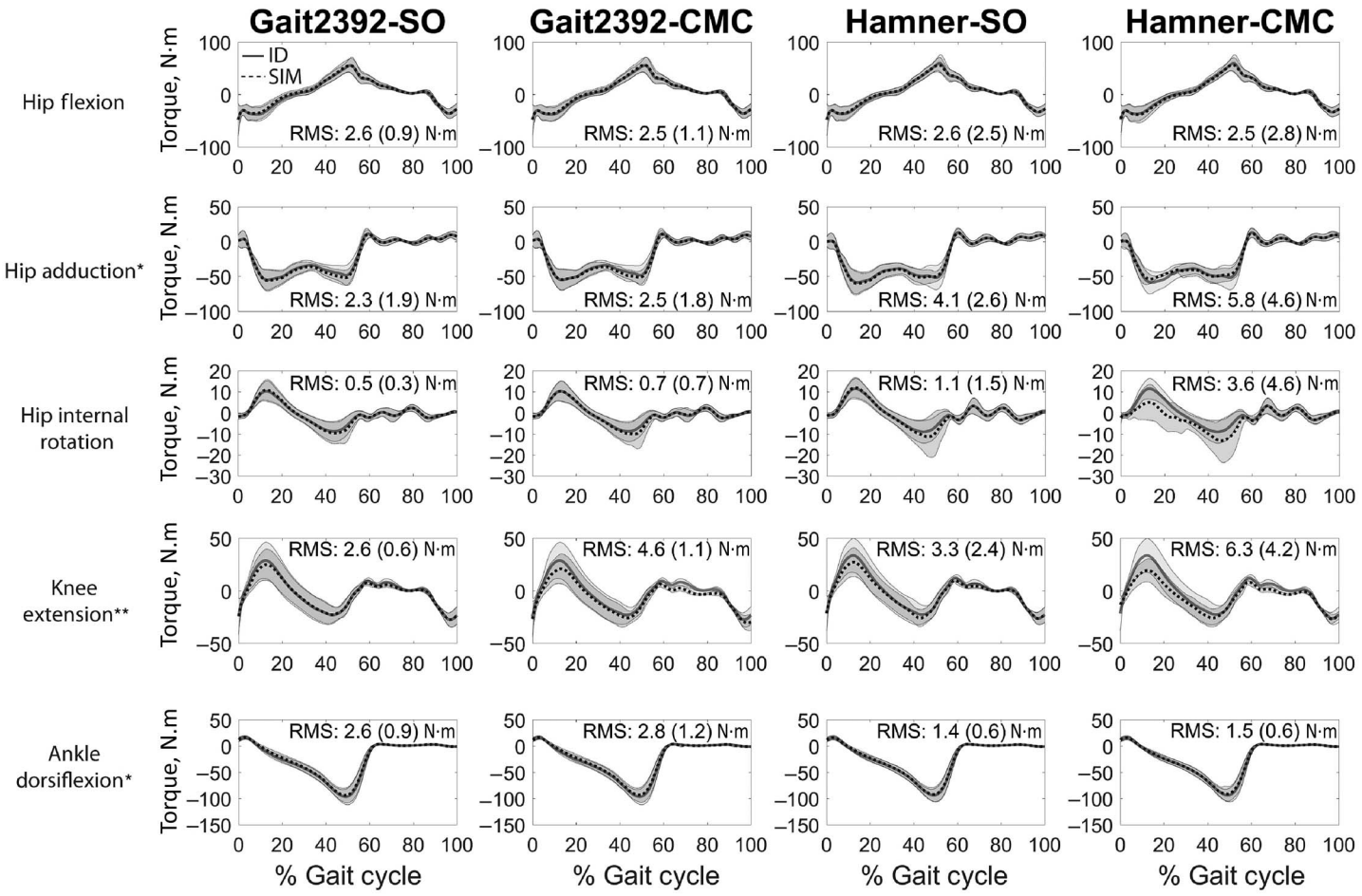

Figure 2 — Average ID- and simulation-derived (SIM) joint torque curves for each condition. RMS errors for each condition are reported within each subplot. Joint torque titles (left) indicate direction of positive torque. Symbols next to joint torque titles indicate significant differences ($P < .05$) between models (*) and optimization techniques (**). CMC indicates computed muscle control; RMS, root mean square; SIM, simulation; SO, static optimization.

simulated muscle activations and forces in young adult gait. Understanding the effects of modeling and optimization assumptions on simulated muscle function and how well the simulation results agree with the experimental data is necessary to draw accurate conclusions from musculoskeletal simulations. In this study, experimental and SO activation pattern agreement was similar between models for all muscles, consistent with the previous findings for these 2 models.[10] However, the agreement between

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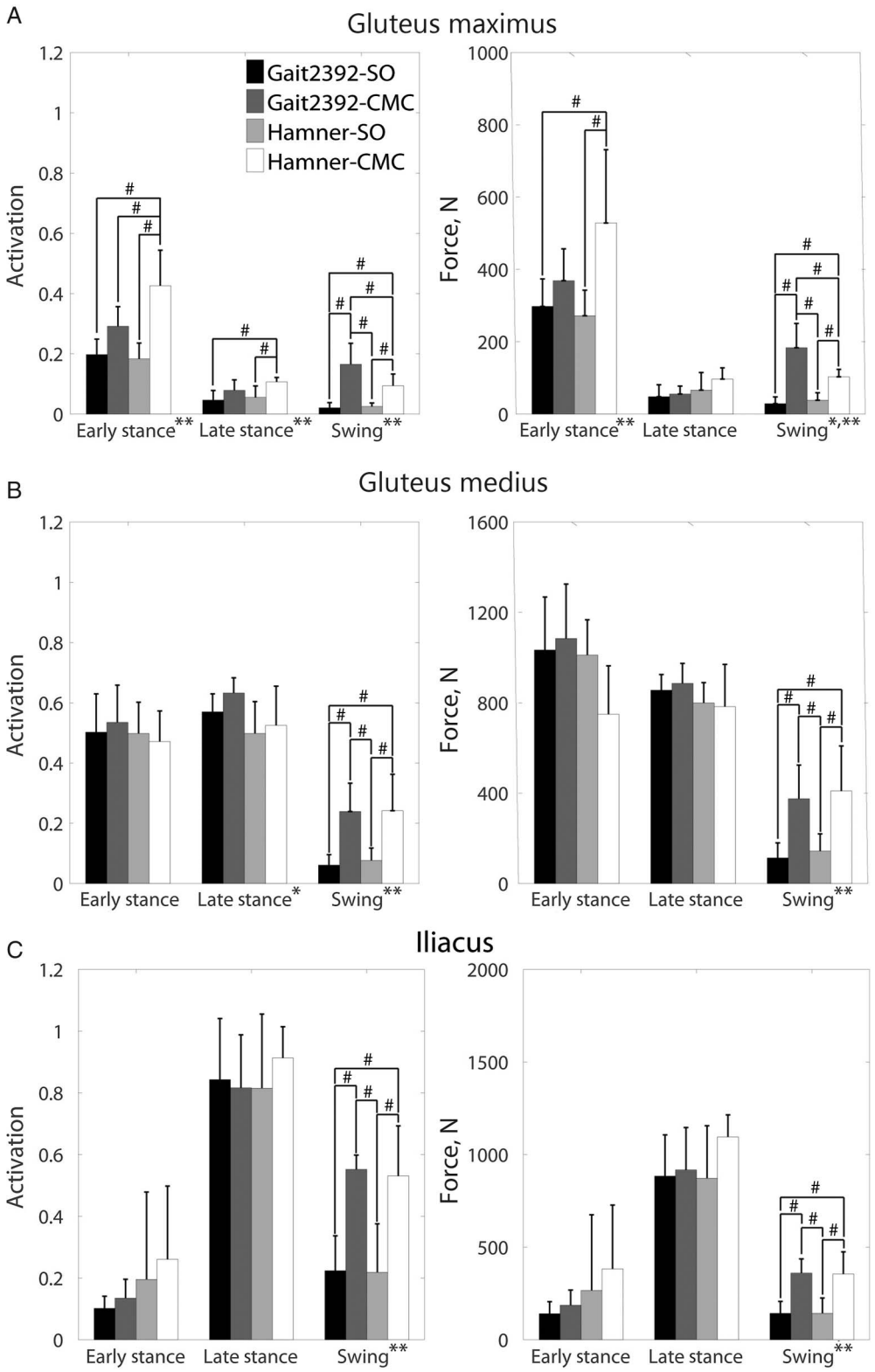

Figure 3 — Average peak muscle activations and forces across gait cycle phase for each condition for (A) gluteus maximus, (B) gluteus medius, and (C) iliacus. Error bars represent 1 SD. Symbols next to gait cycle phase indicate significant differences ($P < .05$) between models (*) and optimization techniques (**). CMC indicates computed muscle control; SO, static optimization. #Indicates pairwise differences between conditions.

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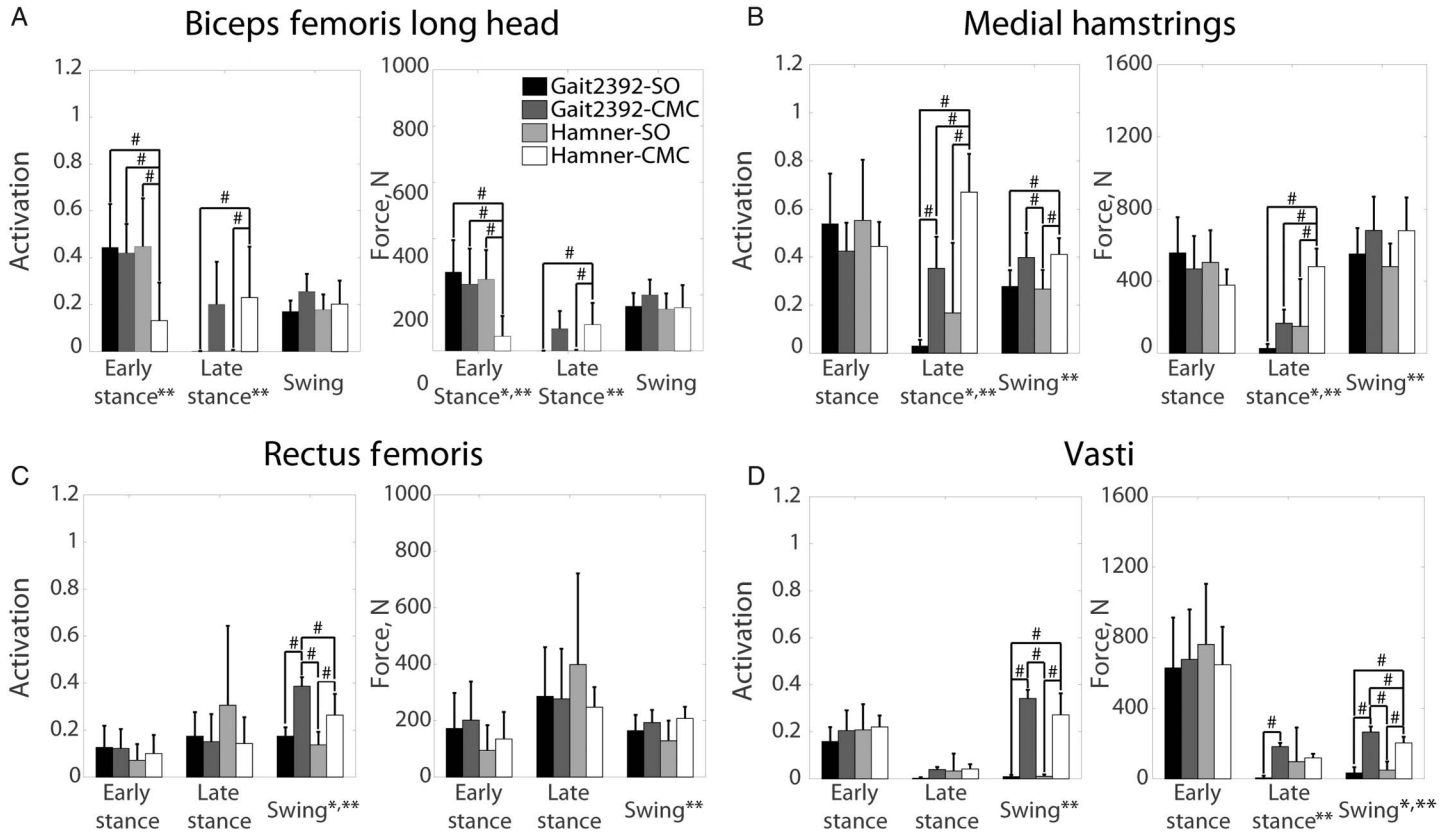

Figure 4 — Average peak muscle activations and forces across gait cycle phase for each condition for (A) biceps femoris long head, (B) medial hamstrings, (C) rectus femoris, and (D) vasti. Error bars represent 1 SD. Symbols next to gait cycle phase indicate significant differences ($P < .05$) between models (*) and optimization techniques (**). CMC indicates computed muscle control; SO, static optimization. #Indicates pairwise differences between conditions.

the CMC and EMG activations was poorer than that of SO with respect to magnitude for tibialis anterior, but better with respect to timing for soleus. Despite the differences in agreement between conditions for individual muscles, there were no differences between conditions for average RMS errors and COS across all 8 muscles, which suggests that the effect of the optimization technique on simulated and experimental activation agreement may be muscle-dependent rather than a fixed characteristic of the optimization technique. For example, Hamner-CMC's higher soleus COS was achieved at the cost of a low-average medial gastrocnemius COS (though not statistically different between conditions). Thus, no single optimization technique and model combination resulted in superior agreement with the experimental measures. Furthermore, because the current literature does not provide a quantitative threshold for an acceptable amount of error between simulated and experimental activations, it is difficult to assess the clinical significance of the RMS errors and COS across conditions in this study. The strong agreement between simulation-derived and ID joint torques indicates that the muscles produced the necessary torques to accurately drive the model's motion. Therefore, the differences between EMG and simulated activations may suggest that neither optimization technique's objective function fully captured the neuromuscular control strategy used by this study's participants. However, muscle model assumptions, including a homogeneous muscle fiber type in the Hill-type muscle model and the use of generic musculotendon parameters, may also contribute to differences between simulated and experimental activation patterns.

Although there were few differences in agreement with the experimental data between conditions, the optimization technique and model choice resulted in significantly different peak activations and forces in all muscles. Consistent with the findings of previous studies,[20,21] CMC generally estimated greater muscle activations and forces than SO, although notable exceptions were observed in biceps femoris long head (Figure 4A) and gastrocnemius (Figure 5A). Moreover, SO activations and forces were less sensitive to model choice than those of CMC. Differences in peak muscle activations and forces between models were observed only with CMC. The only difference between models with SO was the greater tibialis anterior and soleus CCI with Gait2392 compared with Hamner, due to the greater activity of the Gait2392 tibialis anterior during midstance (Figure 1). The greater Gait2392-SO midstance tibialis anterior activity may be partially attributed to the greater ankle range of motion during stance in Gait2392 than Hamner.[10] Due to Gait2392's greater change in ankle angle, its tibialis anterior would experience a larger change in fiber length over the same amount of time as the tibialis anterior of Hamner, leading to greater contraction velocity and lower force-generating capacity. Thus, the Gait2392 tibialis anterior would require greater activation to achieve the same ankle torque as Hamner.[10] However, although the ankle kinematics of Hamner-CMC were the same as those of Hamner-SO, Hamner-CMC also had greater tibialis anterior and soleus CCI than Hamner-SO, which suggests that Hamner is more sensitive to the optimization technique than Gait2392. Furthermore, Hamner-CMC estimated significantly greater soleus late stance force (Figure 5B), significantly smaller biceps femoris early

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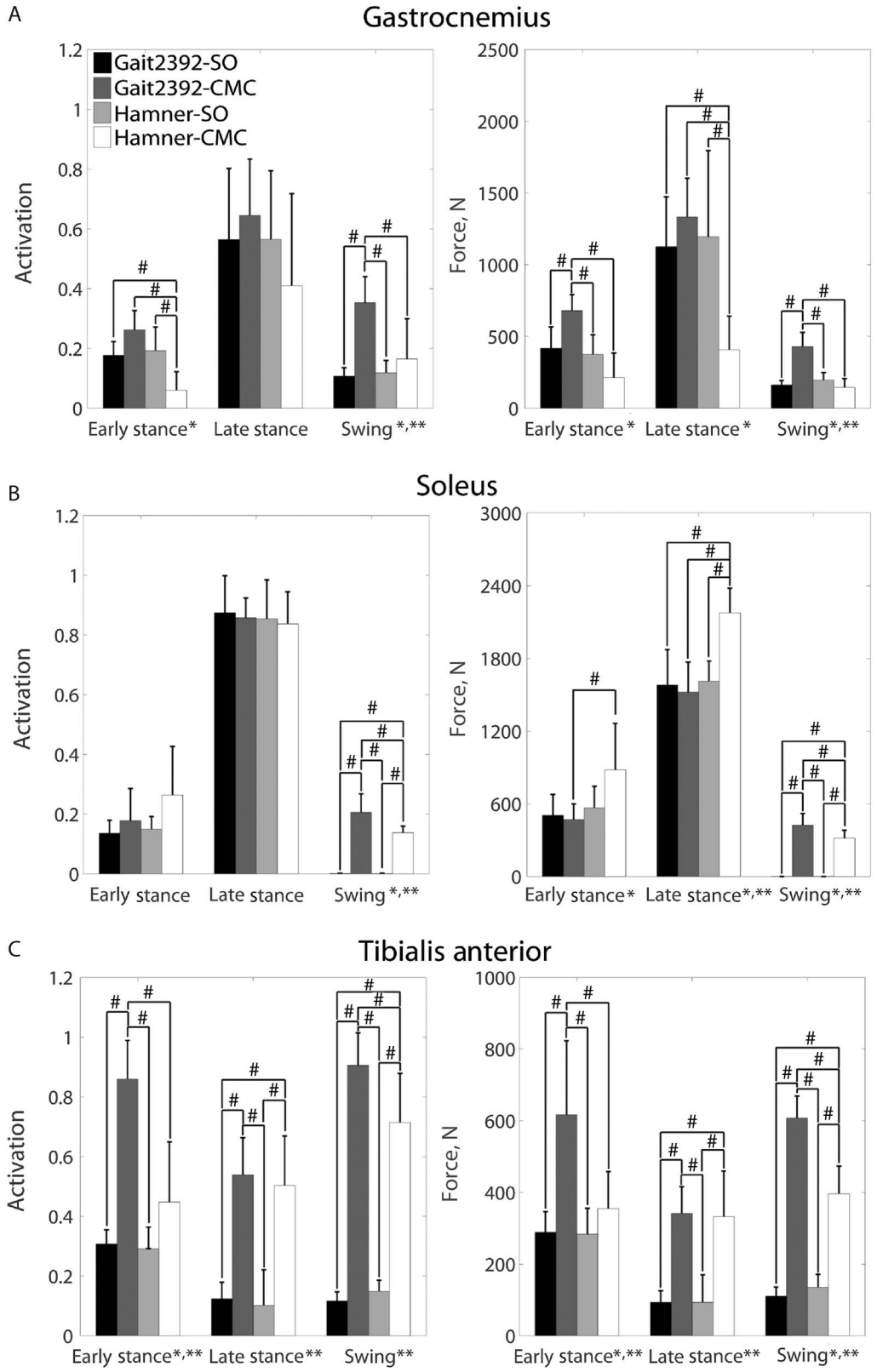

Figure 5 — Average peak muscle activations and forces across gait cycle phase for each condition for (A) gastrocnemius, (B) soleus, and (C) tibialis anterior. Error bars represent 1 SD. Symbols next to gait cycle phase indicate significant differences ($P < .05$) between models (*) and optimization techniques (**). CMC indicates computed muscle control; SO, static optimization. #Indicates pairwise differences between conditions.

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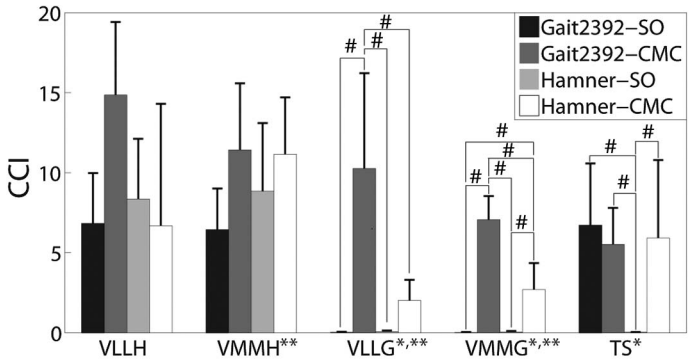

Figure 6 — Average CCI for 5 muscle pairs. Error bars represent 1 SD. Symbols next to CCI name indicate significant differences ($P < .05$) between models (*) and optimization techniques (**). CCI indicates cocontraction indices; CMC, computed muscle control; SO, static optimization; TS, tibialis anterior and soleus; VLLH, lateral vasti and hamstrings; VLLG, lateral vasti and gastrocnemius; VMMG, medial vasti and gastrocnemius; VMMH, medial vasti and hamstrings. #Indicates pairwise differences between conditions.

stance activation and force (Figure 4A), and significantly smaller gastrocnemius early stance activation and late stance force (Figure 5A) than Hamner-SO. However, similar effects of the optimization technique were not observed with Gait2392.

The increased sensitivity of Hamner to the optimization technique is related to Hamner's additional degrees of freedom due to the inclusion of arm segments and the differences in the CMC and SO objective functions used to estimate muscle activations and forces. Hamner's additional coordinates are actuated by torque actuators to track the arm motion. These coordinate actuators are controlled by excitations, which are included in CMC's objective function, along with the muscle excitations. However, the SO objective function only includes muscle activations in its objective function. Therefore, the same number of variables are included in SO and CMC's objective functions with Gait2932. However, with Hamner, CMC's objective function includes more variables than SO's objective function, which may contribute to differences in the estimated muscle activations and forces between optimization techniques.

In addition, the inclusion of passive muscle forces in CMC is a commonly proposed cause of the increased muscle forces observed with CMC compared with SO.[20,21,24] Indeed, a recent study revealed that large differences between CMC and EMG activations were due in part to excessive passive muscle forces.[24] The authors suggested that the excessive passive forces led to a compensatory coactivation of antagonist muscles.[24] However, passive forces contributed minimally to the total force produced by the majority of the muscles investigated in this study (see Appendix Figure A5). Notable exceptions were observed in the rectus femoris and vasti. In Hamner, passive forces contributed up to 65.6% (13.4%) and 94.7% (5.6%) of rectus femoris' total force during late stance and swing, respectively. Similarly, in Gait2392, passive forces contributed up to 65.1% (7.3%) and 84.6% (8.7%) of rectus femoris' total force during late stance and swing, respectively. Thus, the differences in rectus femoris peak activation and force during swing between optimization techniques are explained by these passive forces (Figure 4C). However, the late stance rectus femoris passive forces with CMC did not lead to significant differences between conditions. Passive forces contributed up to 72.1% (16.3%) and 88.4% (6.8%) of vasti's total force during swing in Gait2392 and Hamner, respectively, which explains the greater peak vasti activations and forces observed with CMC compared with SO in both models (Figure 4D). The sensitivity of the CMC rectus femoris and vasti forces to passive force contributions may be an important consideration for studies investigating quadriceps function.

Additional explanations for the differences in muscle function between optimization techniques include the additional constraints of CMC compared with SO, namely, the inclusion of muscle activation-contraction dynamics in CMC and the use of the proportional-derivative feedback controller to calculate the desired accelerations required to accurately track the kinematics. The temporal delays included in CMC to account for muscle activation-contraction dynamics limit the range of forces a muscle can produce at a given time. Based on a muscle's force–length–velocity properties, this constraint also restricts the muscle's excitation range, which will affect the optimization criterion (sum of the squared excitations). The closer agreement between ID- and simulation-derived knee extension torque with SO compared with CMC indicates that the SO muscle forces more accurately reproduced the joint torques driving the experimental motion. The CMC muscle activation-contraction constraints may have limited the torque-generating capacity of the muscles such that the knee extension torques could not be produced.

There are some limitations to this study. The default parameters were used in both SO and CMC; however, the CMC results are sensitive to input parameter values. The authors performed a secondary analysis to assess the effect of 3 CMC input parameters: the look ahead window (LAW; which specifies the allotted time to account for muscle activation-contraction dynamics), the integrator error tolerance, and the maximum number of integrator steps (see Appendix Table A1). Altered parameters had very little effect on agreement between CMC and EMG activations (see Appendix, Figure A6). Only the LAW and error tolerance affected the peak muscle activations and forces (see Appendix, Table A2 and Figures A7–A9). Therefore, changing these parameters may affect the differences between the CMC and SO solutions observed in this study. In addition, the residual and reserve actuator optimal forces were constant across all conditions, there were no constraints on excitations in any condition, and the same tracking task weights were applied in all CMC simulations. Thus, the simulation settings were not optimized to an individual condition, and the results presented are representative of the accuracy of the optimization techniques and model conditions under default settings. Retaining the default settings was necessary to identify the sources of differences in simulated muscle function estimated by different musculoskeletal models and optimization techniques without the confounding effects of different input parameters.

In conclusion, this study revealed that the CMC results are more sensitive to model choice than the SO results and that the Hamner model may be more sensitive to the optimization technique than Gait2392. The results of this study extend the findings of previous work, suggesting that simulation results cannot be validated by published simulation results that used a different model.[10] Therefore, the musculoskeletal model and optimization technique choices should be validated by comparing the simulation results to the experimental data used to generate the simulations, rather than to published data sets.

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Appendix

Condition-Dependent Differences in Muscle Activations and Forces

Gluteus maximus peak activation was greater with CMC than with SO in all phases of the GC ($P \leq .004$). Gluteus maximus had greater peak early stance activation with Hamner-CMC than all other conditions ($P \leq .031$) and greater peak early stance force with Hamner-CMC compared with both SO conditions ($P \leq .018$). Peak early stance gluteus maximus force was also greater with CMC compared with SO ($P = .004$). In late stance, peak gluteus maximus activation was greater with Hamner-CMC compared with both SO conditions ($P \leq .043$), but there were no significant differences in peak late stance force between conditions. In swing, peak gluteus maximus force was greater with CMC than SO ($P < .001$) and greater with Gait2392 than Hamner ($P = .033$). Peak gluteus maximus swing activation and force were greater with Gait2392-CMC than all other conditions (activation: $P \leq .033$; force: $P \leq .007$) and greater with Hamner-CMC than both SO conditions (activation: $P \leq .042$; force: $P \leq .036$).

Gluteus medius peak activation and force were greater with CMC than SO during swing (activation: $P < .001$; force: $P < .001$). Peak gluteus medius late stance activation was greater with Gait2392 than Hamner ($P = .028$); however, there were no significant differences in peak gluteus medius late stance force between conditions ($P \geq .451$). There were also no significant differences between conditions in peak early stance gluteus medius activation or force ($P \geq .060$).

Iliacus peak swing activation and force were greater with CMC than SO (activation: $P < .001$; force: $P < .001$). There were no significant differences in peak early stance or late stance iliacus activations or forces between conditions ($P \geq .329$).

Peak biceps femoris long head early stance activation and force were greater with SO compared with CMC during early stance (activation: $P = .025$; force: $P = .010$). Peak biceps femoris long head early stance forces were greater with Gait2392 than Hamner ($P = .023$). Compared with all other conditions, Hamner-CMC produced smaller peak biceps femoris long head early stance activation ($P \leq .040$) and force ($P \leq .030$). During late stance, biceps femoris long head peak activation and force were greater with CMC than SO (activation: $P = .001$; force: $P < .001$). Hamner-CMC produced greater late stance peak biceps femoris long head activation and force than both SO conditions (activation: $P \leq .049$; force: $P \leq .021$). There were no differences between conditions in peak swing biceps femoris long head activation or force ($P \geq .234$).

Peak medial hamstrings activation and force were greater with CMC than SO in late stance (activation: $P < .001$; force: $P < .001$) and swing (activation: $P < .001$; force: $P = .022$) and greater with Hamner than Gait2392 in late stance (activation: $P = .005$; force: $P = .002$). In late stance, Hamner-CMC produced greater peak medial hamstring activation and forces than all other conditions (activation: $P \leq .029$; force: $P \leq .007$). Gait2392-CMC also produced greater peak late stance medial hamstrings activation than Gait2392-SO ($P = .025$). In swing, Hamner-CMC produced greater peak medial hamstring activation than both SO conditions ($P \leq .044$). There were no differences in early stance peak medial hamstrings activation or force between conditions ($P \geq .284$).

Rectus femoris peak activation and force were greater with CMC than SO during swing (activation: $P < .001$; force: $P = .025$). Gait2392-CMC produced greater peak swing rectus femoris activation than all other conditions ($P \leq .009$), which led to greater swing rectus femoris activation with Gait2392 than with Hamner ($P = .004$). In addition, Hamner-CMC produced greater peak swing rectus femoris activation than Hamner-SO ($P = .007$). There were no differences in early stance or late stance peak rectus femoris activation or force between conditions ($P \geq .386$).

Peak vasti activation and force during swing were greater with CMC than SO (activation: $P < .001$; force: $P < .001$). Gait2392-CMC produced greater peak vasti force during swing compared with all other conditions ($P \leq .049$). Hamner-CMC produced greater peak vasti swing force than both SO conditions (both $P < .001$). Gait2392-CMC produced greater late stance force than Gait2392-SO ($P < .001$), which led to greater late stance vasti force with CMC than with SO ($P = .023$). However, there were no differences in peak late stance vasti activation between conditions ($P \geq .316$). There were also no differences in early stance peak vasti activation or force between conditions ($P \geq .543$).

Tibialis anterior peak activation and force were greater with CMC than SO during early stance (activation: $P < .001$; force: $P < .001$), late stance (activation: $P < .001$; force: $P < .001$), and swing (activation: $P < .001$; force: $P < .001$). Compared with all other conditions, Gait2392-CMC produced greater peak tibialis anterior activation and force during early stance (activations: all $P \leq .001$; forces: $P \leq .008$) and swing (activation: $P \leq .019$; force: all $P < .001$). Following, Gait2392 produced greater peak tibialis anterior early stance activation ($P < .001$) and greater early stance ($P = .016$) and swing ($P < .001$) peak force than Hamner. Hamner-CMC produced greater peak tibialis anterior activation and force during swing compared with both SO conditions (activation: both $P \leq .001$; force: both $P < .001$).

Gastrocnemius' peak early stance activation and force were greater with Gait2392 than Hamner (activation: $P = .002$; force: $P < .001$). Hamner-CMC produced smaller early stance gastrocnemius activation than all other conditions ($P \leq .022$), while Gait2392-CMC produced greater early stance gastrocnemius force than all other conditions ($P \leq .024$). Hamner-CMC produced significantly smaller peak late stance gastrocnemius force compared with all other conditions ($P \leq .022$); however, there were no significant differences in the peak late stance gastrocnemius activation between conditions ($P \geq .370$). Gait2392-CMC produced greater peak swing gastrocnemius activation and force than all other conditions (activation: $P \leq .005$; force: all $p < .001$), which led to greater peak swing gastrocnemius activation and force with CMC than SO (activation: $P < .001$; force: $P < .001$) and with Gait2392 than with Hamner (activation: $P = .018$ force: $P < .001$).

Peak soleus swing activation and force were greater with CMC than SO (activations: $P < .001$; forces: $P < .001$), with Gait2392-CMC peak swing activation and force significantly greater than those of Hamner-CMC (activation: $P = .010$; force: $P = .020$). Hamner-CMC peak soleus force during late stance was greater than with all other conditions ($P \leq .002$), and the early stance Hamner-CMC peak soleus force was greater than that of Gait2392-CMC ($P = .033$). However, there were no significant differences in peak early or late stance soleus activations between conditions ($P \geq .165$).

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Comparison of Experimental and Simulated Muscle Activation Patterns by Condition

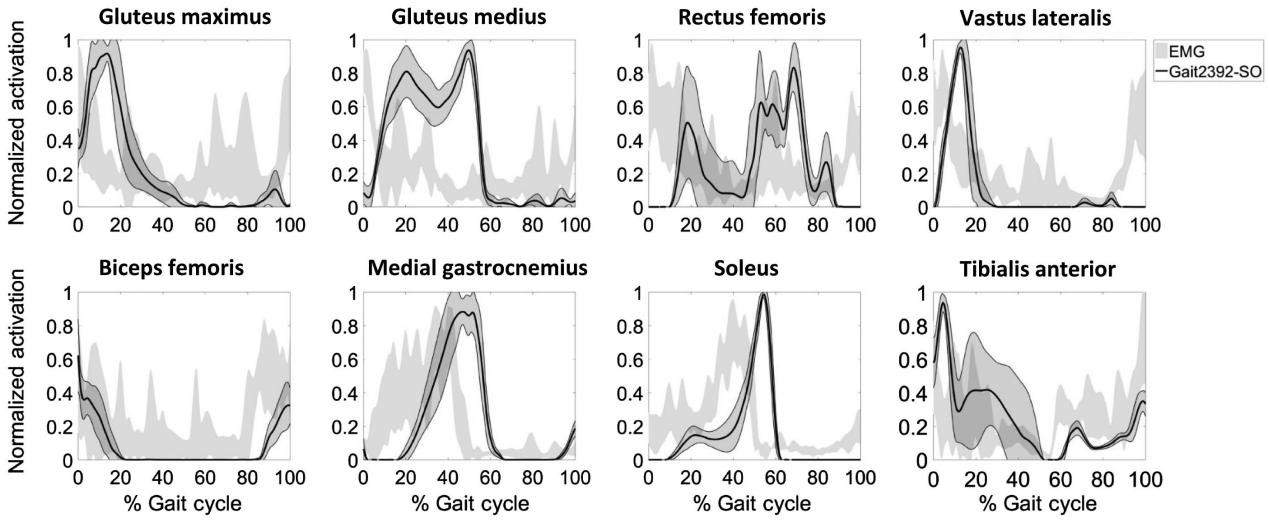

Figure A1 — Normalized EMG and Gait2392-SO muscle activation patterns. Shaded areas represent ±1 SD. Experimental and simulated activations were normalized to the peak activation in the respective trial. EMG indicates electromyography; SO, static optimization.

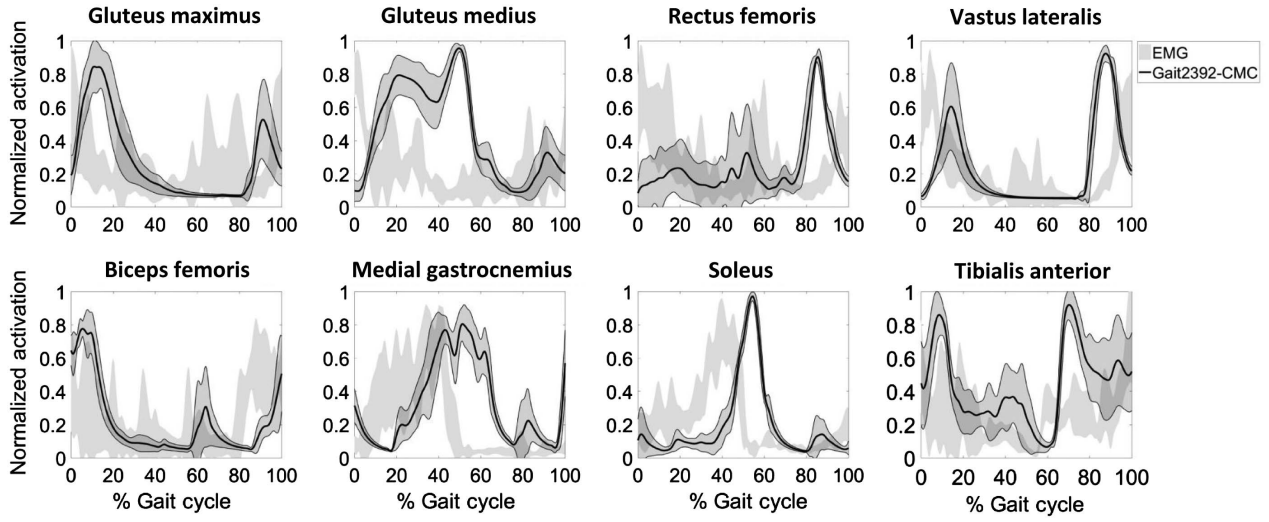

Figure A2 — Normalized EMG and Gait2392-CMC muscle activation patterns. Shaded areas represent ±1 SD. Experimental and simulated activations were normalized to the peak activation in the respective trial. CMC indicates computed muscle control; EMG, electromyography.

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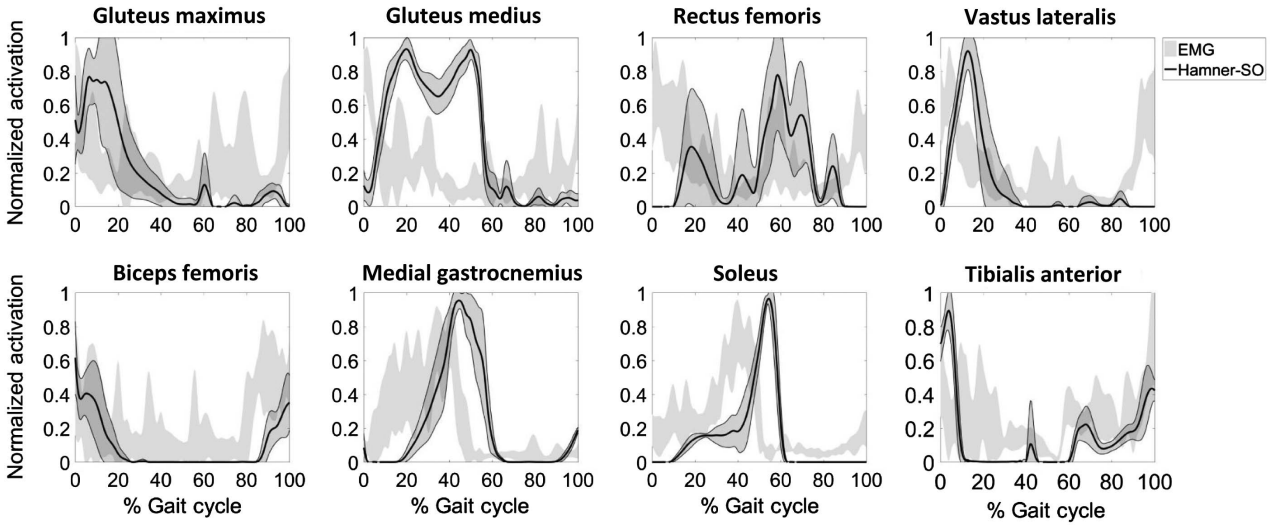

Figure A3 — Normalized EMG and Hamner-SO muscle activation patterns. Shaded areas represent ±1 SD. Experimental and simulated activations were normalized to the peak activation in the respective trial. EMG indicates electromyography; SO, static optimization.

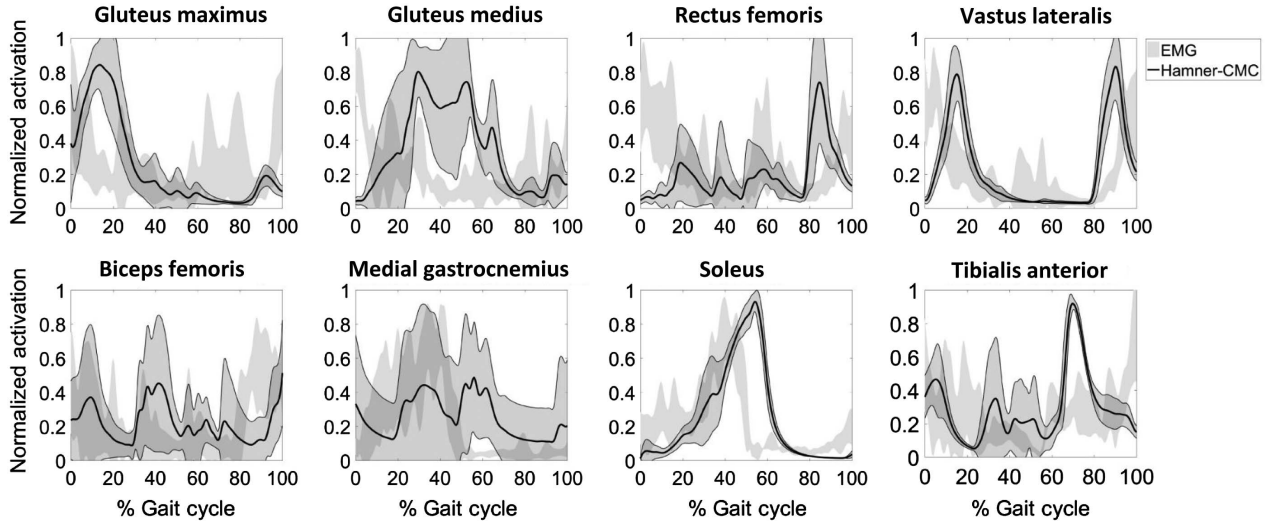

Figure A4 — Normalized EMG and Hamner-CMC muscle activation patterns. Shaded areas represent ±1 SD. Experimental and simulated activations were normalized to the peak activation in the respective trial. CMC indicates computed muscle control; EMG, electromyography.

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Passive Muscle Forces Produced in CMC

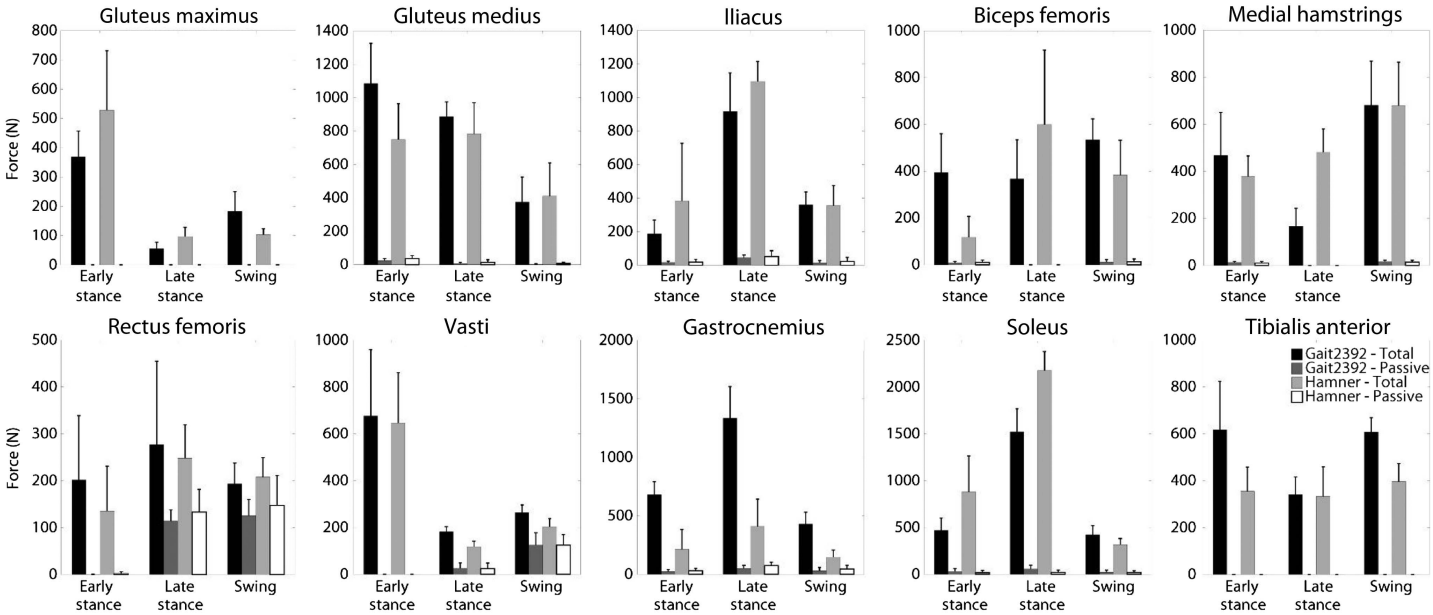

Figure A5 — Average peak total and passive muscle forces from computed muscle control in the Gait2392 and Hamner models. Error bars represent 1 SD.

Effect of Changing CMC Input Parameters on Simulation Results

The authors performed a secondary analysis to assess the effects of changing the values of the CMC LAW, integrator error tolerance, and maximum number of integrator steps on the muscle force and activation results of a single subject in the Gait2392 model. Each parameter value was changed independently, while the other parameters were held at their baseline value (Table A1). The authors compared the RMS, COS, and peak muscle activations and forces by the GC phase of each CMC condition. The subject's Gait2392-SO results are included for reference.

Altered input parameters had little to no effect on agreement between CMC and EMG activations (Figure A6), with no condition's RMS or COS values being different from the baseline value by more than 10.9% (COS of the medial gastrocnemius increased by 0.066 with a 0.02s LAW [COS = 0.0675] compared with the baseline [COS = 0.609]). The LAW and error tolerance, but not the maximum number of integrator steps, impacted peak muscle activations and forces (Table A2). Compared with the baseline LAW (0.01 s), the shorter (0.005s) LAW increased peak activations and forces of the biceps femoris long head (Figure A8–A, late stance and swing), rectus femoris (Figure A8–C, all phases) and, to a lesser extent, the medial hamstrings (Figure A8–B, late stance). The longer (0.02 s) LAW decreased peak activations and forces in the same muscles. Both the shorter and longer LAW decreased the late stance peak activations and forces in the gastrocnemius (Figure A9–A) and tibialis anterior (Figure A9–C). The smaller error tolerance had a less significant impact than the LAW on peak muscle activations and forces, with differences from the baseline of no more than 23.2% (tibialis anterior peak late stance activation, Table A2, Figure A9–C).

Table A1 CMC Input Parameter Sensitivity Analysis: Condition Parameter Values

Condition	Look ahead window, s	Error tolerance	Maximum integrator steps
Baseline	0.01	1×10^{-5}	20,000
Look ahead window = 0.005	0.005	1×10^{-5}	20,000
Look ahead window = 0.02	0.02	1×10^{-5}	20,000
Error tolerance = 1×10^{-6}	0.01	1×10^{-6}	20,000
Max steps = 2K	0.01	1×10^{-5}	2000
Max steps = 200K	0.01	1×10^{-5}	200,000

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Table A2 Percentage Difference From Baseline of Peak Act and Forces for CMC Parameter Conditions[a]

Muscle	Gait cycle phase[b]	LAW = 0.005		LAW = 0.02		Error tolerance = 1×10^{-6}		Max step = 2K		Max step = 200K	
		Act	Force	Act	Force	Act	Force	Act	Force	Act	Force
Gluteus maximus	E Stance	0.3%	1.3%	0.1%	–3.9%	–1.1%	–0.3%	0%	0%	0%	0%
	L Stance	3.1%	3.2%	–3.0%	–2.3%	–0.4%	–0.3%	0%	0%	0%	0%
	Swing	5.6%	6.0%	**−10.3%**	**−10.4%**	–1.0%	–0.6%	0%	0%	0%	0%
Gluteus medius	E Stance	–1.4%	–3.8%	–5.5%	–4.0%	–0.3%	–0.8%	0%	0%	0%	0%
	L Stance	2.3%	0.7%	1.1%	–1.0%	–0.6%	0.2%	0%	0%	0%	0%
	Swing	**19.5%**	**10.5%**	**−11.1%**	–4.1%	0.5%	–0.3%	0%	0%	0%	0%
Iliacus	E Stance	**−13.1%**	–8.0%	**−61.7%**	**−13.0%**	–0.6%	0.0%	0%	0%	0%	0%
	L Stance	**17.0%**	**15.6%**	**14.8%**	**12.8%**	0.0%	2.2%	0%	0%	0%	0%
	Swing	**15.9%**	**23.7%**	**−13.1%**	**−18.8%**	–6.0%	**−10.3%**	0%	0%	0%	0%
Biceps femoris long head	E Stance	0.1%	0.4%	0.7%	2.4%	–0.4%	0.7%	0%	0%	0%	0%
	L Stance	**129.0%**	**117.2%**	**−70.1%**	**−79.8%**	**−21.7%**	**−17.1%**	0%	0%	0%	0%
	Swing	**142.6%**	**50.8%**	**−19.1%**	–6.0%	**−16.9%**	0.1%	0%	0%	0%	0%
Medial hamstrings	E Stance	–0.8%	0.7%	1.9%	2.7%	–0.6%	0.8%	0%	0%	0%	0%
	L Stance	**70.6%**	**65.0%**	**−27.7%**	**−23.7%**	2.9%	3.1%	0%	0%	0%	0%
	Swing	**28.4%**	1.1%	–3.8%	–2.3%	–3.3%	–0.4%	0%	0%	0%	0%
Rectus femoris	E Stance	**51.4%**	**44.5%**	**−24.2%**	**−26.4%**	–2.5%	–1.6%	0%	0%	0%	0%
	L Stance	**188.7%**	**56.9%**	**−48.2%**	**−37.8%**	8.0%	3.7%	0%	0%	0%	0%
	Swing	**20.9%**	**86.4%**	–5.2%	**−42.7%**	0.0%	2.2%	0%	0%	0%	0%
Vasti	E Stance	2.5%	1.9%	–9.3%	**−12.3%**	0.0%	–0.3%	0%	0%	0%	0%
	L Stance	**10.8%**	8.9%	–3.7%	–2.3%	0.2%	–0.6%	0%	0%	0%	0%
	Swing	**21.1%**	6.2%	–7.1%	–1.7%	2.0%	0.0%	0%	0%	0%	0%
Gastrocnemius	E Stance	–2.1%	–0.3%	7.0%	5.9%	–0.4%	0.3%	0%	0%	0%	0%
	L Stance	**−28.9%**	**−14.8%**	**−30.3%**	**−13.5%**	1.2%	–6.4%	0%	0%	0%	0%
	Swing	**17.7%**	**19.7%**	–1.2%	–1.9%	**11.1%**	4.4%	0%	0%	0%	0%
Soleus	E Stance	0.5%	0.7%	–7.6%	–8.0%	–1.7%	–1.3%	0%	0%	0%	0%
	L Stance	–6.3%	–9.8%	–6.8%	**−12.1%**	0.9%	–6.5%	0%	0%	0%	0%
	Swing	**−13.9%**	–6.8%	**−21.9%**	**−16.9%**	**−13.1%**	**−10.2%**	0%	0%	0%	0%
Tibialis anterior	E Stance	2.4%	–0.3%	–2.7%	0.6%	–0.2%	0.0%	0%	0%	0%	0%
	L Stance	**−72.2%**	**−44.3%**	**−76.1%**	**−49.4%**	**−23.2%**	**−18.7%**	0%	0%	0%	0%
	Swing	4.7%	4.9%	–8.9%	**−10.3%**	–2.1%	–3.5%	0%	0%	0%	0%

Abbreviations: ACT, activations; CMC, computed muscle control; E Stance, early stance; L Stance, late stance; LAW, look ahead window; Max, maximum.
[a]Cells with bold values were different from baseline by greater than ±10%.
[b]Gait cycle phases: E Stance, L Stance, and Swing.

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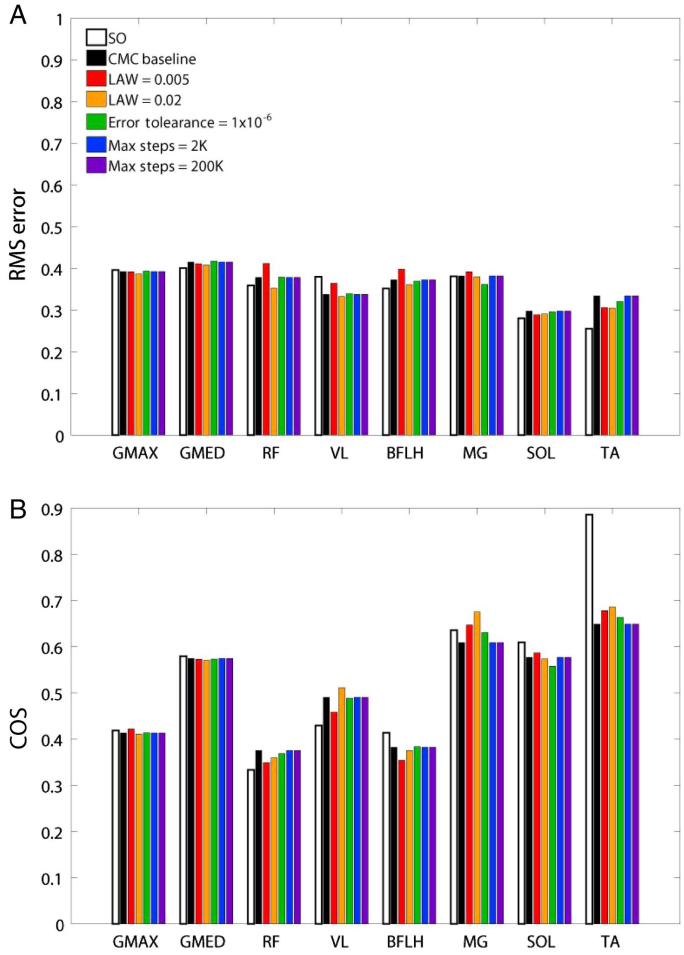

Figure A6 — (A) RMS and (B) COS values for Gait2392-SO and 6 Gait2392-CMC simulations with different CMC input parameters in a representative subject with different input parameter values. The parameters changed were the CMC LAW, integrator error tolerance, and maximum number of integrator steps (max steps). The CMC baseline simulation used the default values for these parameters. BFLH indicates biceps femoris long head; CMC, computed muscle control; COS, cosine of similarity; GMAX, gluteus maximus; GMED, gluteus medius; LAW, look ahead window; MG, medial gastrocnemius; RF, rectus femoris; RMS, root mean square; SO, static optimization; SOL, soleus; TA, tibialis anterior; VL, vastus lateralis.

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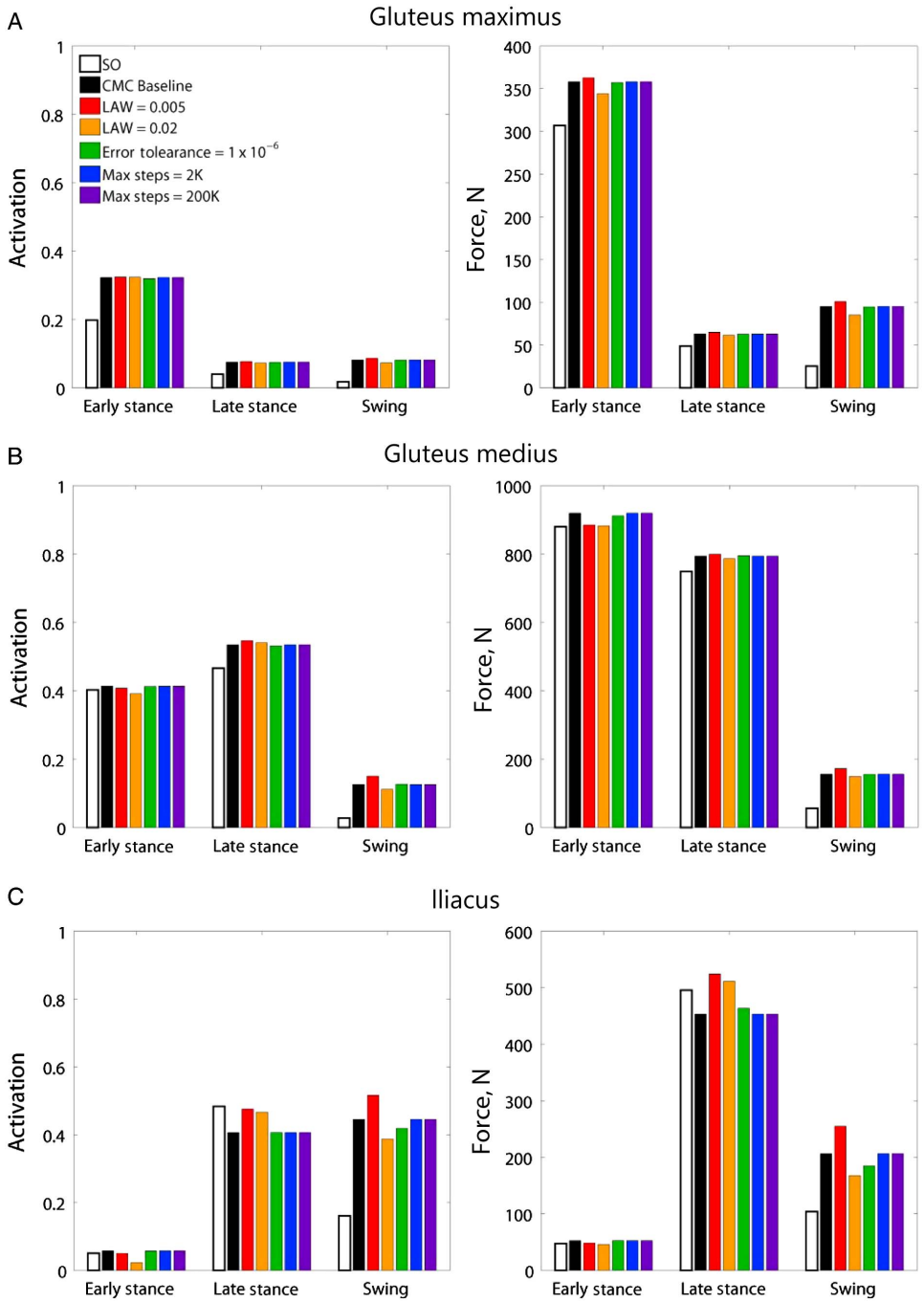

Figure A7 — Peak muscle activations and forces by gait cycle phase for Gait2392-SO and 6 Gait2392-CMC simulations with different CMC input parameters in a representative subject with different input parameter values for (A) gluteus maximus, (B) gluteus medius, and (C) iliacus. The parameters changed were the LAW, integrator error tolerance, and maximum number of integrator steps (max steps). The CMC baseline simulation used the default values for these parameters. CMC indicates computed muscle control; LAW, look ahead window; SO, static optimization.

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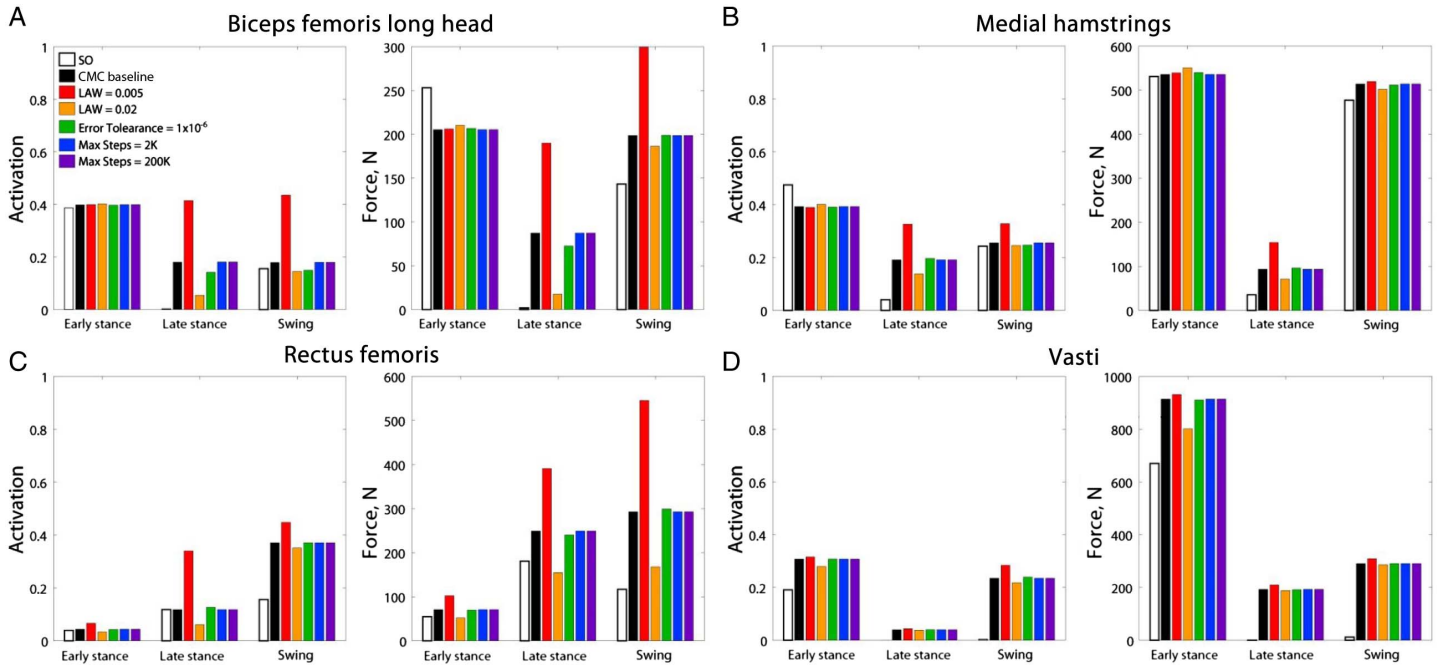

Figure A8 — Peak muscle activations and forces by gait cycle phase for Gait2392-SO and 6 Gait2392-CMC simulations with different input parameters in a representative subject with different CMC input parameter values for the (A) biceps femoris long head, (B) medial hamstrings, (C) rectus femoris, and (D) vasti. The parameters changed were the CMC LAW, integrator error tolerance, and maximum number of integrator steps (max steps). The CMC baseline simulation used the default values for these parameters. CMC indicates computed muscle control; LAW, look ahead window; SO, static optimization.

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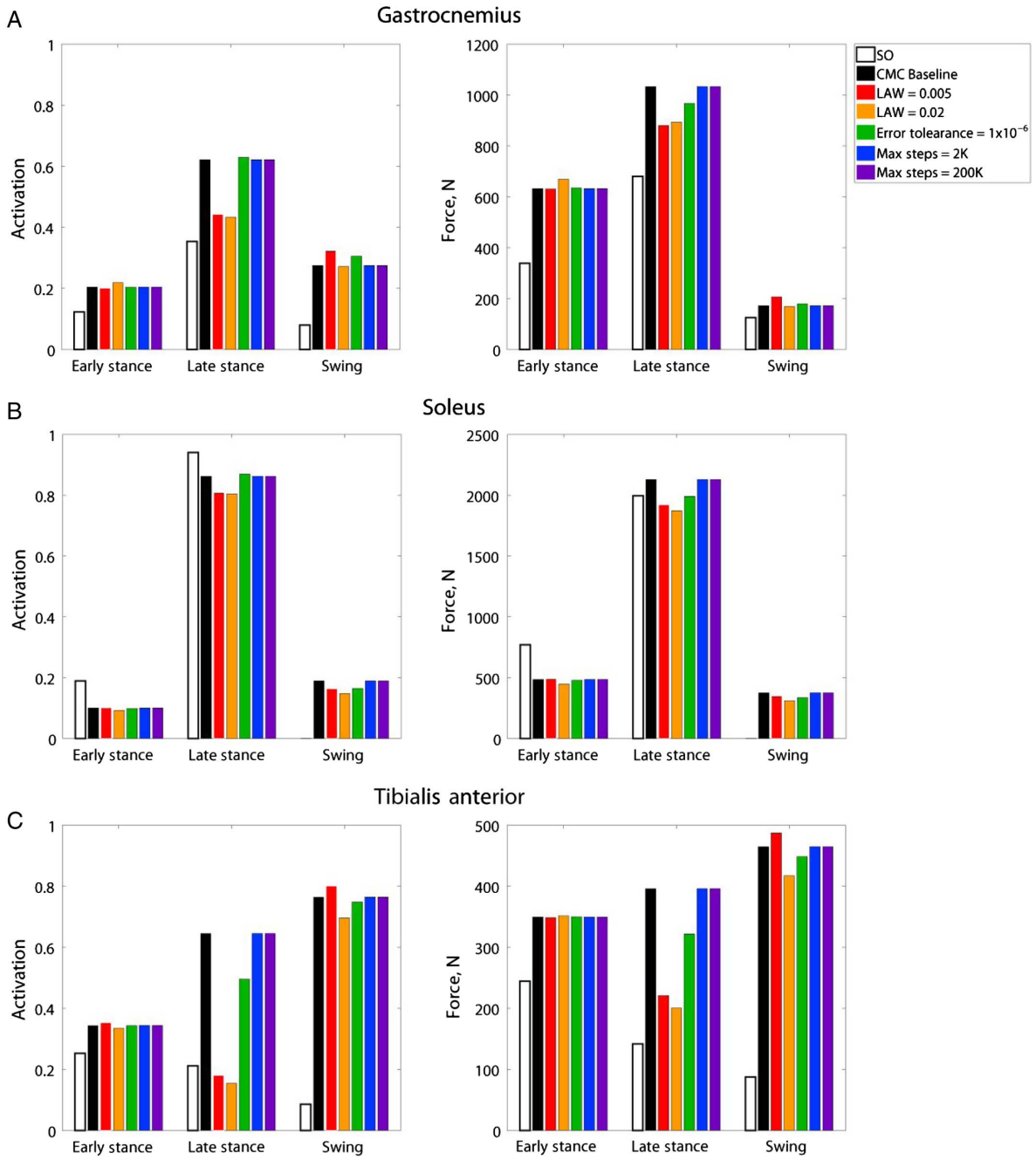

Figure A9 — Peak muscle activations and forces by gait cycle phase for Gait2392-SO and 6 Gait2392-CMC simulations with different CMC input parameters in a representative subject with different input parameter values for (A) gastrocnemius, (B) soleus, and (C) tibialis anterior. The parameters changed were the CMC LAW, integrator error tolerance, and maximum number of integrator steps (max steps). The CMC baseline simulation used the default values for these parameters. CMC indicates computed muscle control; LAW, look ahead window; SO, static optimization.

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